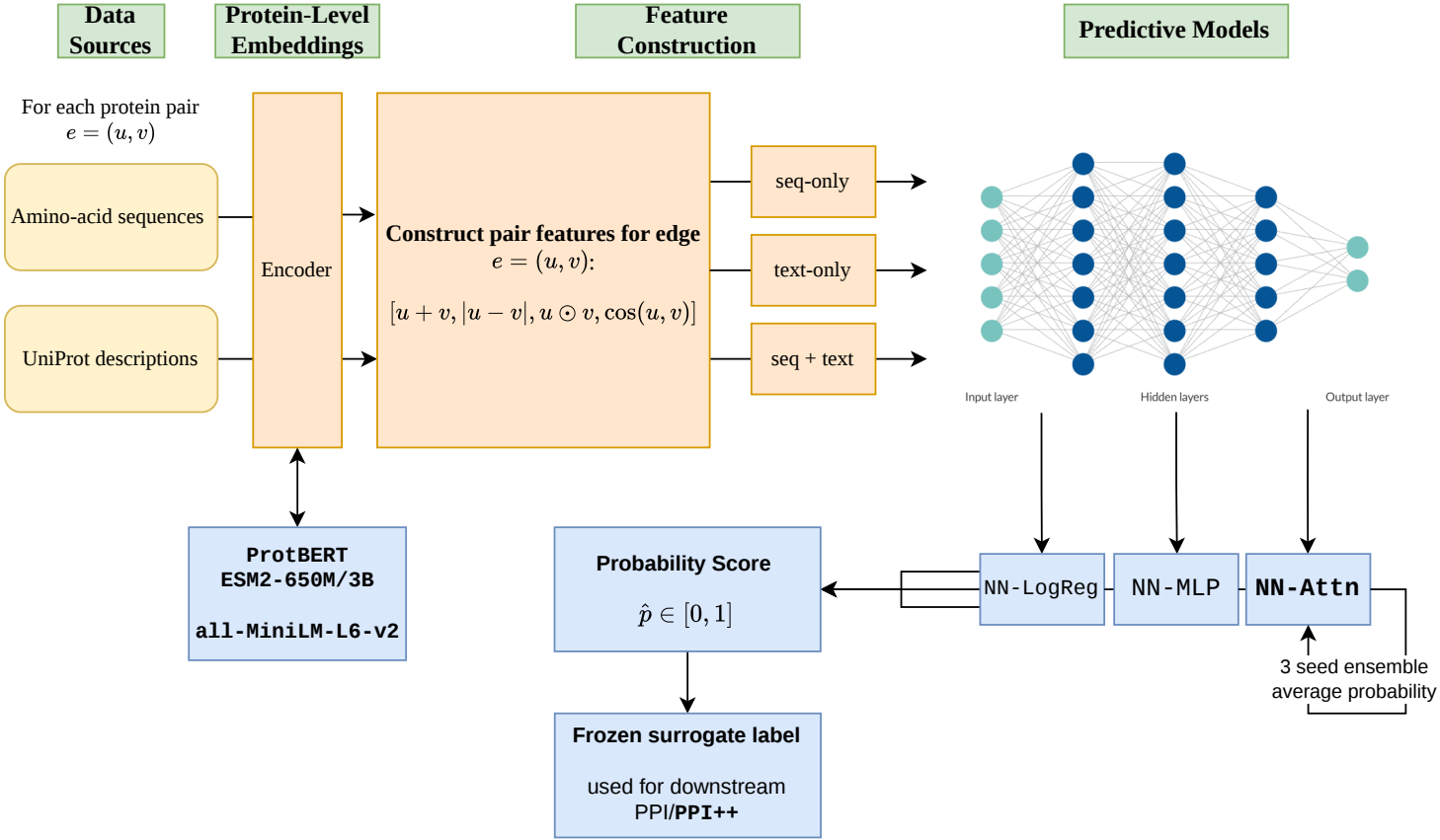
Data Sources
Protein-Level Embeddings
Feature Construction
Predictive Models
For each protein pair $e = (u, v)$
Amino-acid sequences
UniProt descriptions
Encoder
Construct pair features for edge $e = (u, v)$:
$[u + v, |u - v|, u \odot v, \cos(u, v)]$
seq-only
text-only
seq + text
Input layer
Hidden layers
Output layer
ProtBERT
ESM2-650M/3B
all-MiniLM-L6-v2
Probability Score
$\hat{p} \in [0, 1]$
NN-LogReg
NN-MLP
NN-Attn
3 seed ensemble
average probability
Frozen surrogate label
used for downstream
PPI/PPI++